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Problem 1

Context-Free Grammar I

(1) Language $L = \{\, wcw^R \mid w \in \{a,b\}^*,\ |w| \text{ even} \,\}$.

A parity-tracking mirror grammar (start S_0 = even):

$$S_0 \to aS_1a \mid bS_1b \mid c \qquad S_1 \to aS_0a \mid bS_0b.$$

(2) Language $L = \{\, a^i b^j c^k \mid i,j,k \geq 1,\ i+k = j \,\}$.

$$S \to AB, \qquad A \to aAb \mid ab, \qquad B \to bBc \mid bc.$$

Problem 2

Context-Free Grammar II and BNF/EBNF

(1) $L = \{\, w^p x^{2n} y^{3n} z^m \mid p > m,\ m,n \geq 1 \,\}$.

(a) Shortest string: $m = 3, n = 1, p = 4$. `wwwwxxyyyzzz`.

(b) CFG:

$$S \to wSz \mid B$$
$$B \to wwVRz$$
$$V \to wV \mid \varepsilon$$
$$R \to xxRyyy \mid xxyyy$$

(2) BNF $\to$ EBNF:

$$\langle \textit{func_call} \rangle ::= \texttt{id "(" [} \langle \textit{expr} \rangle \texttt{ \{"," } \langle \textit{expr} \rangle \texttt{\}] ")"}.$$

(3) EBNF $\to$ BNF:

$$\langle \textit{declaration} \rangle ::= \langle \textit{opt_static} \rangle\ \langle \textit{type} \rangle\ \langle \textit{id} \rangle\ \langle \textit{id_list} \rangle$$
$$\langle \textit{opt_static} \rangle ::= \texttt{static} \mid \varepsilon$$
$$\langle \textit{id_list} \rangle ::= \texttt{","}\ \langle \textit{id} \rangle\ \langle \textit{id_list} \rangle \mid \varepsilon.$$

Problem 3

Ambiguity

(1) Grammar with dangling else is ambiguous. Unambiguous form:

$$\begin{aligned}
\langle stmt \rangle &::= \langle matched \rangle \mid \langle unmatched \rangle \\
\langle matched \rangle &::= \texttt{if } \langle expr \rangle \texttt{ then } \langle matched \rangle \texttt{ else } \langle matched \rangle \mid \langle other_stmt \rangle \\
\langle unmatched \rangle &::= \texttt{if } \langle expr \rangle \texttt{ then } \langle stmt \rangle \mid \texttt{if } \langle expr \rangle \texttt{ then } \langle matched \rangle \texttt{ else } \langle unmatched \rangle.
\end{aligned}$$

(2) Ambiguous grammar for expressions. Unambiguous version (precedence $\wedge > * > +$):

$$\begin{aligned}
E &\to E + T \mid T \\
T &\to T * G \mid G \\
G &\to H \ \wedge \ G \mid H \\
H &\to \texttt{id} \mid (E).
\end{aligned}$$

Problem 4

Ambiguity and Derivation

(1) Grammar $S \to$ `<tag>`S`</tag>` | `text` | SS. Derive `<b>hello</b><i>world</i>`.

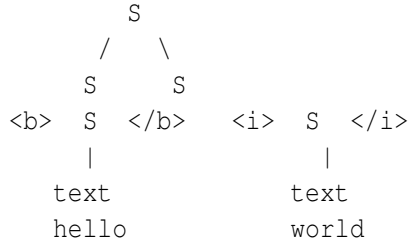
Leftmost derivation:

$S \Rightarrow SS$
$\Rightarrow$ `<b>`S`</b>`S
$\Rightarrow$ `<b>text</b>`S
$\Rightarrow$ `<b>hello</b><i>`S`</i>`
$\Rightarrow$ `<b>hello</b><i>text</i>`
$\Rightarrow$ `<b>hello</b><i>world</i>`

Rightmost derivation:

$S \Rightarrow SS$
$\Rightarrow S$`<i>`S`</i>`
$\Rightarrow S$`<i>text</i>`
$\Rightarrow S$`<i>world</i>`
$\Rightarrow$ `<b>`S`</b><i>world</i>`
$\Rightarrow$ `<b>text</b><i>world</i>`
$\Rightarrow$ `<b>hello</b><i>world</i>`

(ASCII) parse tree:



(2a) Two different derivations aren't enough to prove ambiguity; you need two different leftmost (or two different rightmost) derivations.

(2b) Grammar

$$S \to AC \mid DB, \quad A \to aAb \mid \varepsilon, \quad C \to cC \mid \varepsilon, \quad D \to aD \mid \varepsilon, \quad B \to bBc \mid \varepsilon$$

is ambiguous: `abc` can be derived in two distinct ways.

Problem 5

Regular Expression to CFG

(1) Identifier `[A-Za-z_][A-Za-z0-9_]*`.

$$
\begin{aligned}
ID &\to HEAD\ REST \\
HEAD &\to a \mid \cdots \mid z \mid A \mid \cdots \mid Z \mid _ \\
REST &\to CHAR\ REST \mid \varepsilon \\
CHAR &\to HEAD \mid DIGIT \\
DIGIT &\to 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9.
\end{aligned}
$$

(2) URI `(http|https)://(www.)?[A-Za-z0-9]+.(com|org|edu).`

$$
\begin{aligned}
URI &\to PROTO\ \texttt{"://"}\ OPTWWW\ NAME\ \texttt{"."}\ SUFFIX \\
PROTO &\to \texttt{"http"} \mid \texttt{"https"} \\
OPTWWW &\to \texttt{"www."} \mid \varepsilon \\
NAME &\to ALNUM\ NAME \mid ALNUM \\
ALNUM &\to a \mid \cdots \mid z \mid A \mid \cdots \mid Z \mid 0 \mid \cdots \mid 9 \\
SUFFIX &\to \texttt{"com"} \mid \texttt{"org"} \mid \texttt{"edu"}.
\end{aligned}
$$